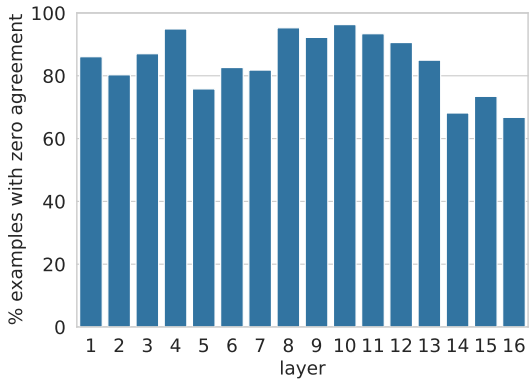

Figure 8: The fraction of examples in a random sample of 4,000 examples where the layer's prediction is different from the prediction of all of its memories.

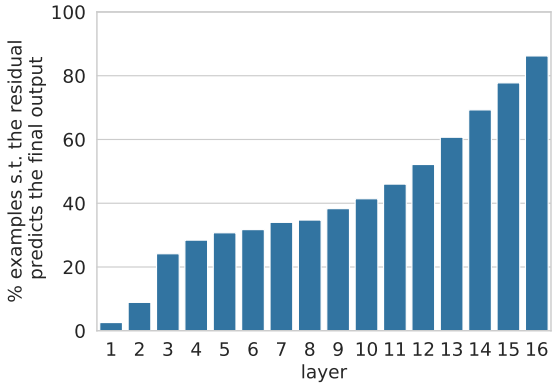

Figure 9: Fraction of examples in each layer, where the residual's top prediction matches the model's output.

be "cached" in individual memory cells, and do not require compositionality.

5.2 Inter-Layer Prediction Refinement

While a single feed-forward layer composes its memories in parallel, a multi-layer model uses the residual connection $\mathbf{r}$ to *sequentially* compose predictions to produce the model's final output:[5]

$$\mathbf{x}^\ell = \text{LayerNorm}(\mathbf{r}^\ell)$$
$$\mathbf{y}^\ell = \text{FF}(\mathbf{x}^\ell)$$
$$\mathbf{o}^\ell = \mathbf{y}^\ell + \mathbf{r}^\ell$$

We hypothesize that the model uses the sequential composition apparatus as a means to *refine* its prediction from layer to layer, often deciding what the prediction will be at one of the lower layers.

To test our hypothesis, we first measure how often the probability distribution induced by the residual vector $\mathbf{r}^\ell$ matches the model's final output $\mathbf{o}^L$ (L being the total number of layers):

$$\text{top}(\mathbf{r}^\ell) = \text{top}(\mathbf{o}^L)$$

Figure 9 shows that roughly a third of the model's predictions are determined in the bottom few layers. This number grows rapidly from layer 10 onwards, implying that the majority of "hard" decisions occur before the final layer.

We also measure the probability mass p that each layer's residual vector $\mathbf{r}^\ell$ assigns to the model's final prediction:

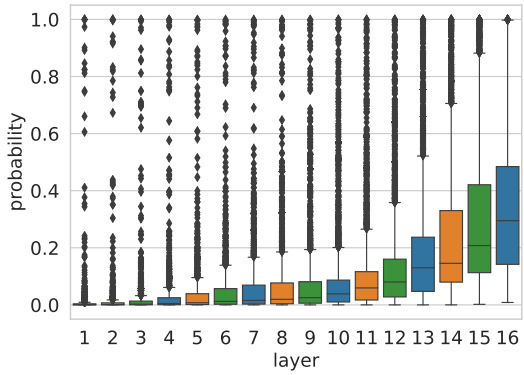

Figure 10: Probability of the token output by the model according to the residual of each layer.

$$w = \text{top}(\mathbf{o}^L)$$
$$\mathbf{p} = \text{softmax}(\mathbf{r}^\ell \cdot E)$$
$$p = \mathbf{p}_w$$

Figure 10 shows a similar trend, but emphasizes that it is not only the top prediction's identity that is refined as we progress through the layers, it is also the model's confidence in its decision.

To better understand how the refinement process works at each layer, we measure how often the residual's top prediction changes following its interaction with the feed-forward layer ($\text{top}(\mathbf{r}^\ell) \neq \text{top}(\mathbf{o}^\ell)$), and whether this change results from the feed-forward layer overriding the residual ($\text{top}(\mathbf{o}^\ell) = \text{top}(\mathbf{y}^\ell)$) or from a true composition ($\text{top}(\mathbf{r}^\ell) \neq \text{top}(\mathbf{o}^\ell) \neq \text{top}(\mathbf{y}^\ell)$).

Figure 11 shows the breakdown of different cases per layer. In the vast majority of examples, the residual's top prediction ends up being the

[5]The residual propagates information from previous layers, including the transformer's self-attention layers.

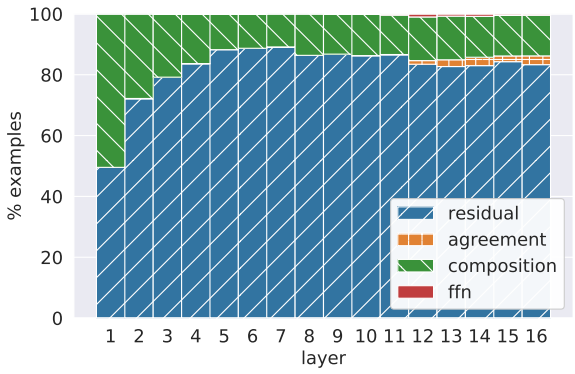

Figure 11: Breakdown of examples by prediction cases: the layer's output prediction matches the residual's prediction (*residual*), matches the feed-forward layer's prediction (*ffn*), matches both of the predictions (*agreement*), or none of the predictions (*composition*). By construction, there are no cases where the residual's prediction matches the feed-forward layer's prediction and does not match the output's prediction.

model's prediction (*residual+agreement*). In most of these cases, the feed forward layer predicts something different (*residual*). Perhaps surprisingly, when the residual's prediction does change (*composition+ffn*), it rarely changes to the feed-forward layer's prediction (*ffn*). Instead, we observe that composing the residual's distribution with that of the feed-forward layer produces a "compromise" prediction, which is equal to neither (*composition*). This behavior is similar to the intra-layer composition we observe in Section 5.1. A possible conjecture is that the feed-forward layer acts as an elimination mechanism to "veto" the top prediction in the residual, and thus shifts probability mass towards one of the other candidate predictions in the head of the residual's distribution.

Finally, we manually analyze 100 random cases of last-layer composition, where the feed-forward layer modifies the residual output in the *final* layer. We find that in most cases (66 examples), the output changes to a semantically distant word (e.g., *"people"* → *"same"*) and in the rest of the cases (34 examples), the feed-forward layer's output shifts the residual prediction to a related word (e.g. *"later"* → *"earlier"* and *"gastric"* → *"stomach"*). This suggests that feed-forward layers tune the residual predictions at varying granularity, even in the last layer of the model.

6 Related Work

Considerable attention has been given to demystifying the operation of neural NLP models. An extensive line of work targeted neuron functionality in general, extracting the properties that neurons and subsets of neurons capture (Durrani et al., 2020; Dalvi et al., 2019; Rethmeier et al., 2020; Mu and Andreas, 2020; Vig et al., 2020), regardless of the model architecture or neurons' position in it. Jacovi et al. (2018) analyzed CNN architectures in text classification and showed that they extract key n-grams from the inputs.

The study of the transformer architecture has focused on the role and function of self-attention layers (Voita et al., 2019; Clark et al., 2019; Vig and Belinkov, 2019) and on inter-layer differences (i.e. lower vs. upper layers) (Tenney et al., 2019; Jawahar et al., 2019). Previous work also highlighted the importance of feed-forward layers in transformers (Press et al., 2020; Pulugundla et al., 2021; Xu et al., 2020). Still, to date, the role of feed-forward layers remains under-explored.

Also related are interpretability methods that explain predictions (Han et al., 2020; Wiegreffe and Pinter, 2019), however, our focus is entirely different: we do not interpret individual predictions, but aim to understand the mechanism of transformers.

Characterizing the functionality of memory cells based on examples that trigger maximal activations has been used previously in NLP (Rethmeier et al., 2020) and vision (Erhan et al., 2009).

7 Discussion and Conclusion

Understanding how and why transformers work is crucial to many aspects of modern NLP, including model interpretability, data security, and development of better models. Feed-forward layers account for most of a transformer's parameters, yet little is known about their function in the network.

In this work, we propose that feed-forward layers emulate key-value memories, and provide a set of experiments showing that: (a) keys are correlated with human-interpretable input patterns; (b) values, mostly in the model's upper layers, induce distributions over the output vocabulary that correlate with the next-token distribution of patterns in the corresponding key; and (c) the model's output is formed via an aggregation of these distributions, whereby they are first composed to form individual layer outputs, which are then refined throughout the model's layers using residual connections.

Our findings open important research directions:

- **Layer embedding space.** We observe a correlation between value distributions over the output

vocabulary and key patterns, that increases from lower to upper layers (Section 4). Is this because the layer's output space transforms across layers? If so, how? We note that this possible transformation cannot be explained solely by the function of feed-forward layers: if the model only did a series of key-value look-ups and value-distribution aggregation via weighted addition, then a single, unifying embedding space would appear more natural. Thus, the transformation might have to do with the interplay between feed-forward layers and self-attention layers.

- **Beyond language modeling.** Our formulation of feed-forward networks as key-value memories generalizes to any transformer model, e.g. BERT encoders and neural translation models. We thus expect our qualitative empirical observations to hold across diverse settings, and leave verification of this for future work.

- **Practical implications.** A better understanding of feed-forward layers has many implications in NLP. For example, future studies may offer interpretability methods by automating the pattern-identification process; memory cells might affect training-data privacy as they could facilitate white-box membership inference (Nasr et al., 2019); and studying cases where a correct pattern is identified but then suppressed during aggregation may guide architectural novelties.

Thus, by illuminating the role of feed-forward layers, we move towards a better understanding of the inner workings of transformers, and open new research threads on modern NLP models.

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